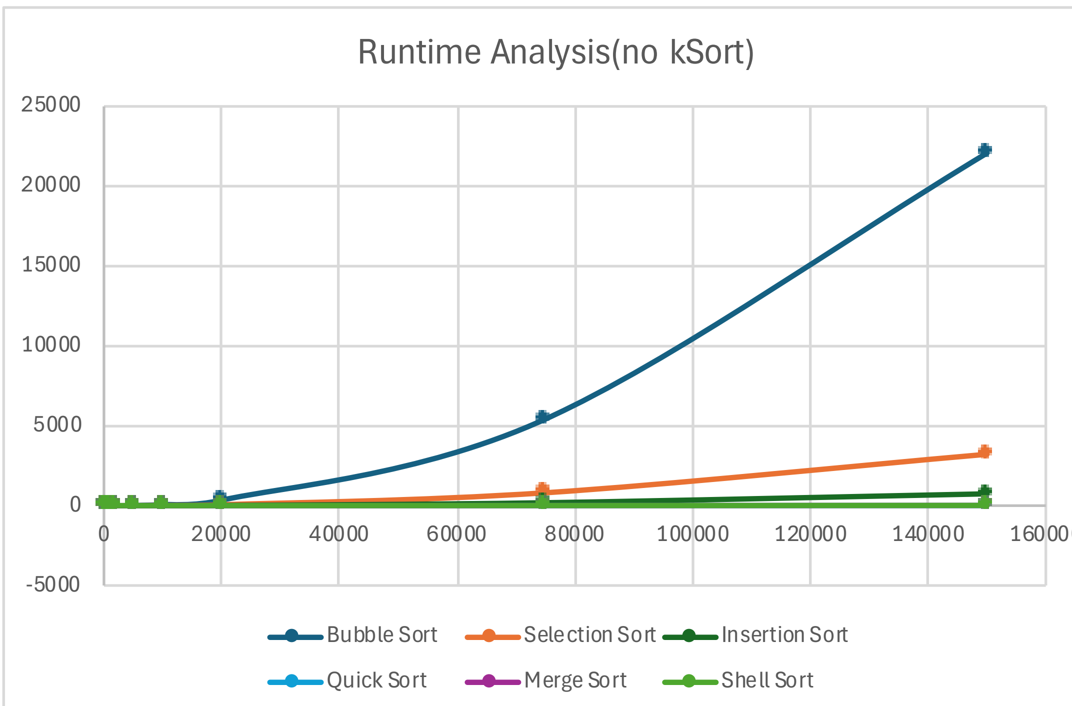
Sizes	Bubble Sort	Selection Sort	Insertion Sort	Quick Sort	Merge Sort	Shell Sort
100	0	0	0	0	0	0
500	0	0	0	0	0	0
1000	0	0	0	0	0	0
2000	1	0	0	0	0	0
5000	10.1	3.05	0	0	0	0
10000	69.2	14.05	2.25	0	0	0
20000	340.55	57.85	11.45	0	1	1
75000	5421.9	806.7	182.45	3	5.3	5.85
150000	22076.85	3217.1	740.1	7.1	11	13

My initial predictions were mostly corect. Bubble sort was by far the most ineffiecent even though it has a same worst case as both insertion and selection sort. One part of the my prediction that was proven wrong was that insertion sort was considerably faster than selection sort. Originally I had thought that it would be pretty similar but, insertion eneded being much more efficent. Another part of my prediction that was proven correct was that shell, quick, and merge sort were so similar in there times. The sorting algorithms are defiently two tiered. The relatively simple algorithms such as insertion, bubble, and selection are much slower than the more complicated ones (merge, quick, shell). This is



Runtime Analysis(no kSort)
25000
20000
15000
10000
5000
0
-5000
0
20000
40000
60000
80000
100000
120000
140000
Bubble Sort
Selection Sort
Insertion Sort
Quick Sort
Merge Sort
Shell Sort

0